In [ ]:
```
# EXPERIMENT 3 – CREATING DATAFRAMES & ANALYZING PROPERTIES
PIN NO.: 241UDS0103

AIM
To create Pandas DataFrames using NumPy arrays and CSV files, and analyze
DataFrame properties using structured renewable-energy data..
```

In [ ]:
```
## Step 1: Install and Import Libraries
```

In [1]:
```
import numpy as np
import pandas as pd
```

In [ ]:
```
## Step 2: Create a 2-D Array
```

In [2]:
```
energy_output_241UDS0103 = np.array([
    [420, 455, 510, 535],
    [380, 405, 438, 462],
    [295, 310, 328, 346]
])
energy_output_241UDS0103
```

Out[2]:
```
array([[420, 455, 510, 535],
       [380, 405, 438, 462],
       [295, 310, 328, 346]])
```

In [3]:
```
energy_output_241UDS0103.shape, energy_output_241UDS0103.ndim
```

Out[3]:
```
((3, 4), 2)
```

In [ ]:
```
## Step 3: Convert the 2-D Array into a DataFrame
```

In [4]:
```
energy_index_241UDS0103 = ["Solar", "Wind", "Hydro"]
energy_columns_241UDS0103 = ["Week_1", "Week_2", "Week_3", "Week_4"]
energy_df_241UDS0103 = pd.DataFrame(
    data=energy_output_241UDS0103,
    index=energy_index_241UDS0103,
    columns=energy_columns_241UDS0103
)
energy_df_241UDS0103
```

Out[4]:

| | Week_1 | Week_2 | Week_3 | Week_4 |
|---|---|---|---|---|
| **Solar** | 420 | 455 | 510 | 535 |
| **Wind** | 380 | 405 | 438 | 462 |
| **Hydro** | 295 | 310 | 328 | 346 |

In [5]:
```
energy_df_241UDS0103["Week_5"] = [552, 480, 360]
energy_df_241UDS0103
```

Out[5]:

| | Week_1 | Week_2 | Week_3 | Week_4 | Week_5 |
|---|---|---|---|---|---|
| **Solar** | 420 | 455 | 510 | 535 | 552 |
| **Wind** | 380 | 405 | 438 | 462 | 480 |
| **Hydro** | 295 | 310 | 328 | 346 | 360 |

In [6]:
```
new_source_241UDS0103 = pd.DataFrame(
    {"Week_1": [220], "Week_2": [235], "Week_3": [248],
     "Week_4": [260], "Week_5": [275]},
    index=["Geothermal"]
)
energy_df_241UDS0103 = pd.concat([energy_df_241UDS0103,
new_source_241UDS0103])
energy_df_241UDS0103
```

Out[6]:

| | Week_1 | Week_2 | Week_3 | Week_4 | Week_5 |
|---|---|---|---|---|---|
| **Solar** | 420 | 455 | 510 | 535 | 552 |
| **Wind** | 380 | 405 | 438 | 462 | 480 |
| **Hydro** | 295 | 310 | 328 | 346 | 360 |
| **Geothermal** | 220 | 235 | 248 | 260 | 275 |

In [ ]:
```
## Step 4: Create a DataFrame from an Existing CSV File
```

In [7]:
```
energy_data_241UDS0103 =
pd.read_csv("Experiment_3_241UDS0103_Renewable_Energy.csv")
energy_data_241UDS0103
```

Out[7]:

| | PIN | Energy_Source | Month | Energy_Output_MWh | Efficiency_Percent | Maintenance_Hours |
|---|---|---|---|---|---|---|
| **0** | 241UDS0103 | Solar | January | 420 | 86 | 18 |
| **1** | 241UDS0103 | Solar | February | 455 | 91 | 20 |
| **2** | 241UDS0103 | Solar | March | 510 | 95 | 22 |
| **3** | 241UDS0103 | Wind | January | 380 | 79 | 24 |
| **4** | 241UDS0103 | Wind | February | 405 | 83 | 26 |
| **5** | 241UDS0103 | Wind | March | 438 | 88 | 28 |
| **6** | 241UDS0103 | Hydro | January | 295 | 72 | 15 |
| **7** | 241UDS0103 | Hydro | February | 310 | 75 | 16 |
| **8** | 241UDS0103 | Hydro | March | 328 | 78 | 17 |

In [ ]:
```
## Step 5: Analyze the Important Properties of the DataFrame
```

In [8]:
```
energy_data_241UDS0103.head()
```

Out[8]:

| | PIN | Energy_Source | Month | Energy_Output_MWh | Efficiency_Percent | Maintenance_Hours |
|---|---|---|---|---|---|---|
| **0** | 241UDS0103 | Solar | January | 420 | 86 | 18 |
| **1** | 241UDS0103 | Solar | February | 455 | 91 | 20 |
| **2** | 241UDS0103 | Solar | March | 510 | 95 | 22 |
| **3** | 241UDS0103 | Wind | January | 380 | 79 | 24 |
| **4** | 241UDS0103 | Wind | February | 405 | 83 | 26 |

In [9]:
```
energy_data_241UDS0103.shape
```

Out[9]:
```
(9, 6)
```

In [10]:
```
energy_data_241UDS0103.index
```

Out[10]:
```
RangeIndex(start=0, stop=9, step=1)
```

In [11]:
```
energy_data_241UDS0103.dtypes
```

Out[11]:
```
PIN                   str
Energy_Source         str
Month                 str
Energy_Output_MWh     int64
Efficiency_Percent    int64
Maintenance_Hours     int64
dtype: object
```

In [12]:
```
energy_data_241UDS0103.columns
```

Out[12]:
```
Index(['PIN', 'Energy_Source', 'Month', 'Energy_Output_MWh',
       'Efficiency_Percent', 'Maintenance_Hours'],
      dtype='str')
```

In [ ]:
```
## Step 6: Information About the DataFrame
```

In [13]:
```
energy_data_241UDS0103.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 6 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   PIN                 9 non-null      str
 1   Energy_Source       9 non-null      str
 2   Month               9 non-null      str
 3   Energy_Output_MWh   9 non-null      int64
 4   Efficiency_Percent  9 non-null      int64
 5   Maintenance_Hours   9 non-null      int64
dtypes: int64(3), str(3)
memory usage: 564.0 bytes
```

In [ ]:
```
## Step 7: Descriptive Statistics
```

In [14]:
```
energy_data_241UDS0103.describe()
```

Out[14]:

| | Energy_Output_MWh | Efficiency_Percent | Maintenance_Hours |
|---|---|---|---|
| **count** | 9.000000 | 9.000000 | 9.000000 |
| **mean** | 393.444444 | 83.000000 | 20.666667 |
| **std** | 71.864649 | 7.648529 | 4.609772 |
| **min** | 295.000000 | 72.000000 | 15.000000 |
| **25%** | 328.000000 | 78.000000 | 17.000000 |
| **50%** | 405.000000 | 83.000000 | 20.000000 |
| **75%** | 438.000000 | 88.000000 | 24.000000 |
| **max** | 510.000000 | 95.000000 | 28.000000 |

In [ ]:
```
## Step 8: Transpose Rows and Columns
```

In [15]:
```
energy_data_241UDS0103.head(10).transpose()
```

Out[15]:

Columns 1–8 of 10

| | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| **PIN** | 241UDS0103 | 241UDS0103 | 241UDS0103 | 241UDS0103 | 241UDS0103 | 241UDS0103 | 241UDS0103 |
| **Energy_Source** | Solar | Solar | Solar | Wind | Wind | Wind | Hydro |
| **Month** | January | February | March | January | February | March | January |
| **Energy_Output_MWh** | 420 | 455 | 510 | 380 | 405 | 438 | 295 |
| **Efficiency_Percent** | 86 | 91 | 95 | 79 | 83 | 88 | 72 |
| **Maintenance_Hours** | 18 | 20 | 22 | 24 | 26 | 28 | 15 |

Columns 9–10 of 10 · first column repeated

| | 7 | 8 |
|---|---|---|
| **PIN** | 241UDS0103 | 241UDS0103 |
| **Energy_Source** | Hydro | Hydro |
| **Month** | February | March |
| **Energy_Output_MWh** | 310 | 328 |
| **Efficiency_Percent** | 75 | 78 |
| **Maintenance_Hours** | 16 | 17 |

In [ ]:
```
RESULT
NumPy and Pandas were used to create a DataFrame from a 2-D array, add a
column and row, load a CSV file, and inspect DataFrame properties,
information, descriptive statistics, and transposed data.
```